

Homework 8

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Exercise 1

Let X_i and Y be binary variables, with $P(X_i = 1|Y = 1) = \beta_{1i}$ and $P(X_i = 0|Y = 1) = 1 - \beta_{1i}$. If $x_i = 1$:

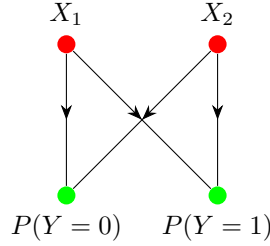
$$\beta_{1i}x_i + (1 - \beta_{1i})(1 - x_i) = \beta_{1i} \cdot 1 + (1 - \beta_{1i})(1 - 1) = \beta_{1i}$$

Instead, if $x_i = 0$:

$$\beta_{1i}x_i + (1 - \beta_{1i})(1 - x_i) = \beta_{1i} \cdot 0 + (1 - \beta_{1i})(1 - 0) = 1 - \beta_{1i}$$

In either case, $P(X_i = x_i|Y = 1) = \beta_{1i}x_i + (1 - \beta_{1i})(1 - x_i)$.

Exercise 2



Consider the sigmoid function, which maps the real line to $(0, 1)$:

$$\begin{aligned}
 P(Y = 1|X_1 = x_1, X_2 = x_2) &= \frac{1}{1 + e^{-(w_1x_1 + w_2x_2 + b)}} \\
 P(Y = 0|X_1 = x_1, X_2 = x_2) &= 1 - P(Y = 1|X_1 = x_1, X_2 = x_2) \\
 &= 1 - \frac{1}{1 + e^{-(w_1x_1 + w_2x_2 + b)}} \\
 &= \frac{e^{-(w_1x_1 + w_2x_2 + b)}}{1 + e^{-(w_1x_1 + w_2x_2 + b)}} \\
 &= \frac{1}{1 + e^{(w_1x_1 + w_2x_2 + b)}}
 \end{aligned}$$

Now consider the hyperbolic tangent function, which maps the real line to $(-1, 1)$:

$$f(x_1, x_2) = \tanh(e^{w_1x_1 + x_2x_2 + b})$$

This modified function maps the real line to $(-1, 1)$:

$$f(x_1, x_2) = \frac{1 + \tanh(e^{w_1x_1 + x_2x_2 + b})}{2}$$

Then, letting $z = w_1x_1 + x_2x_2 + b$, note that:

$$\begin{aligned}
 \frac{1 + \tanh(e^z)}{2} &= \frac{1 + \frac{e^{2z} - 1}{e^{2z} + 1}}{2} \\
 &= \frac{\frac{2e^{2z}}{e^{2z} + 1}}{2} = \frac{e^{2z}}{e^{2z} + 1} \\
 &= \frac{1}{1 + e^{-2z}}
 \end{aligned}$$

We specify the structural equations for the endogenous variables $P(Y = 0)$ and $P(Y = 1)$:

$$\begin{aligned}
 P(Y = 0) &= \frac{1 + \tanh(e^{w_{0,1}x_1 + x_{0,2}x_2 + b_0})}{2} = \frac{1}{1 + e^{-2(w_{0,1}x_1 + x_{0,2}x_2 + b_0)}} \\
 P(Y = 1) &= \frac{1 + \tanh(e^{w_{1,1}x_1 + x_{1,2}x_2 + b_1})}{2} = \frac{1}{1 + e^{-2(w_{1,1}x_1 + x_{1,2}x_2 + b_1)}}
 \end{aligned}$$

with parameters $w_{0,1}, w_{0,2}, b_0, w_{1,1}, w_{1,2}, b_1$. To reproduce the logistic model, we set:

$$w_{0,1} = -\frac{w_1}{2}, w_{0,2} = -\frac{w_2}{2}, b_0 = -\frac{b}{2}$$
$$w_{1,1} = \frac{w_1}{2}, w_{1,2} = \frac{w_2}{2}, b_1 = \frac{b}{2}$$

Exercise 3

The measures of lack of fit discussed are:

Sum of squared errors: If the data points are (x_i, y_i) and the model predictions are $f(x_i)$, the sum of squared errors is $\sum_i (y_i - f(x_i))^2$, and measures the deviation from the data points.

Number of misclassifications: For binary classification, the number of misclassifications is the sum of the number of false positives and false negatives.

Sum of negative log likelihoods: If the data points are (x_i, y_i) , the sum of negative log likelihoods is $\sum_i -\log P(Y = y_i | X = x_i)$, and measures the probability of deviating from the data points.